

CHAPTER 8

**ACCEPTANCE OF E-HEALTH BY
HEALTHCARE PROFESSIONALS IN
DEVELOPING COUNTRIES: CASE OF
MOROCCO**

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8.1 INTRODUCTION

Healthcare organizations include medical and administrative units. Medical ones deploy skills and knowledge to promulgate patient care. As for the administrative units, they provide necessary tools to support medical activities. Both of these two units generate diverse information. Thus, usage of Information Technology (IT) becomes a crucial strategic challenge for these organizations (Ammenwerth et al., 2004). Some authors noted that IT usage offers opportunities ranging from cost reduction and clinical errors, support of care professionals and improving the quality of care efficiency (Scott et al., 2007).

Since 1997, Morocco, as a developing country, has undergone several issues related to financing arrangement, care access, productivity and quality of healthcare organizations. These issues led the Moroccan government to a process of structural reforms enabling health sectors to improve management processes, quality and performance of healthcare organizations.

In 2001, the government launched an integrated project to support healthcare reforms: the Project Management and Financing of Health Sector (PMFHS). This aimed to introduce managerial tools to support changes in healthcare system. It provided the Moroccan population with integrated and standardized care services. It gave better support to healthcare professionals and to improve the quality and efficiency of the promulgated care services. Implementing IT in the administrative and healthcare processes was considered as one of the main goals of this project. Despite these efforts to enhance the care system, the project still faces to two major shortcomings: (i) an unequal distribution of health care promulgation, and (ii) an inadequate hospital management.

In 2005, Moroccan government launched E-health project. This later aimed to provide timely reliable indicators to health decision makers and to improve availability and access to information and data related to citizens and health professionals. Moreover, a strategy 2008–2012 was proposed, it aimed ensuring fairness of care offer between regions and at facilitating access to the rural population. It also aimed to provide citizens confidence toward health system by reducing hospital cost and improving health quality.

Furthermore, another strategy is proposed for the period 2012–2016. It is considered as a consolidation and a continuum of the past strategies. Improvement of the health system governance was the pillar axe of this strategy. It is focused on IT usage and its contribution to improve health quality within healthcare organizations.

As part of this initiative, Provincial Hospital (PH) Hassan Ist of Tiznit city, located in south of Morocco expected to integrate IT within its departments. Many objectives are behind this integration: improve the care quality and communication, reduce waiting time for patients, and reduce hospital costs. However, this integration is impeded by to cultural barriers related to the reluctance of health professionals (nurses and physicians) toward IT acceptance.

In fact, acceptance and integration of e-health into the everyday life of healthcare workers became a reality in developing as well as developed countries (World Health Organisation, 2004). It is almost impossible to make a strategic decision without involving this technology in healthcare organizations (Arlotto and Oakes, 2003). Hence, healthcare organizations invested heavily in IT expecting to improve their service quality and performance (Davaraj and Kohli, 2003).

Yet, the expected benefits cannot be harvested unless this technology is accepted by health professionals. Thus, the main objective of this chapter is to identify factors predicting E-health acceptance by physicians and nurses working in developing countries. It suggests a model based on Technology Acceptance Model (TAM) (Davis et al., 1989) extended by two added constructs “Image” and “Trust.” TAM is considered a salient model that was widely used to predict IT acceptance in many contexts; nevertheless, it is not sufficiently used to predict Healthcare information technologies in developing countries. Hence, determined factors will help stakeholders involved in E-health and healthcare information systems projects especially in developing countries.

The study was conducted in the Provincial Hospital (PH) Hassan Ist of Tiznit city, located in south of Morocco. The choice of this hospital is due essentially to: (1) It belongs to a region considered as the pilot site for the E-health integration in Morocco (2) This hospital is considered as the second organization in the region that plans to integrate E-health.

The rest of the chapter is organized as follows: In the theoretical background section we review relevant theories and models of technology acceptance, we add studies based on TAM to predict E-health acceptance. Specific hypotheses are developed in the research model and Hypothesis Development section. The methodology section describes the quantitative study and the results are presented in the results section. We then go over our results in the discussion section. The chapter ends with a brief conclusion.

8.2 THEORIES AND MODELS PREDICTING INFORMATION TECHNOLOGY ACCEPTANCE

Research in information systems studied why and how individuals accept IT (Venkatesh et al., 2003). It refers to theories and models that deal with IT acceptance issue (Kukafka et al., 2007). Many theoretical models attempt to predict the level of IT acceptance and even explain why and how this technology is used. The following section reviews briefly these theories and describes TAM model considered as the basis to build up a framework allowing prediction of IT acceptance.

8.2.1 THEORIES PREDICTING INDIVIDUAL BEHAVIOR TOWARD INFORMATION TECHNOLOGY

The behavioral approach related to acceptance and usage of technologies provides a general framework to understand and predict individual behavior toward IT. It mobilizes theories from several fields: Sociology, Psychology and Management Information Systems. The main objective of this subsection is to review the relevant theories predicting IT acceptance by individuals.

8.2.1.1 Innovation Diffusion Theory

In 1983, Rogers developed Innovation Diffusion Theory (IDT) to understand diffusion and adoption of innovations. This theory provides a conceptual framework to identify factors influencing IT acceptance by individuals and organizations. It describes the process through which new ideas, practices, or technologies are spread into a social system (Rogers, 2003).

In the beginning, IDT attempted to understand innovation adoption by a social group only. Furthermore, several studies showed that it could be used for individual and collective decisions toward IT adoption (Compeau et al., 1999). In 1995, Rogers identified five factors predicting diffusion innovation adoption: (1) perceived attributes; (2) type of decision (3) communication channel; (4) social system; and (5) change agent. He suggested that to accept an innovation, the individual should first be informed of its existence. After then, he could develop an attitude towards this innovation, and then decide whether to accept or reject it. Hence, an individual with a positive attitude towards an innovation should, first, adopt it and then use it. Rogers states that the perceived attributes include five theoretical constructs to accept a new technology.

1. Relative Advantage: Degree to which an innovation is perceived as being better than the idea it supersedes. It is not necessary if an innovation has more objective advantages. What actually matters is whether and individual perceived this innovation as being beneficial (advantageous).

2. Compatibility: Degree to which an innovation is perceived as consistent with existing values, style of work, past experiences, social practices and standards users. An innovation that is incompatible with the values and norms of a social system will not be accepted as rapidly as the one that is compatible.

3. Complexity: Degree to which an innovation is perceived as difficult to use. New ideas that are simple to understand will be accepted much faster than others that require development of new skills for their comprehension.

4. Trialability: Degree to which an innovation may be experimented with a limited basis. The opportunity to test an innovation allows potential users to have more confidence; and,

5. Observability: Degree to which results and benefits of innovation are visible and accessible. Individuals are more likely to accept an innovation whenever its results are clear.

Furthermore, accepted innovations are those that have a strong comparative advantage, an ability to try, a satisfactory visibility, a low complexity, and finally a high compatibility.

IDT aims to understand the phenomenon of innovations acceptance (Rogers, 1995), however it has many limits: It does not explain clearly IT acceptance, or how to avoid users reluctances toward this acceptance (Chau and Tam, 1997). Hence the need for other theories that would predict technology acceptance.

8.2.1.2 Theory of Reasoned Action

In 1975, Fishbein and Ajzen developed Theory of Reasoned Action (TRA). It is a psychosocial theory that explains behavior in an interaction perspective between individual and his social environment. It provides a solid basis for understanding and predicting the individual behavior (Benbasat and Barki, 1995). It proposes a model predicting individual intention. This model suggests that individual behavior is directly determined by Behavioral Intention that is influenced by attitude and subjective norms.

Unlike the IDT based on the innovation itself, the TRA examines perceptions of the actual use of an innovation (Moore and Benbassat, 1991). It is seen as a general model that can be used for predicting individual behavior. Many limits of this theory could be listed: The first one concerned the relationship between attitudes and subjective norms. Miniard and Cohen (1981) noted that the TRA did not distinguish effectively between attitude and subjective norm impacts on Intention. As for the second limit, it is due to the fact that the behavior is under individual control, something that is highly questionable.

Faced with these limits, Ajzen (1991) modified the TRA and proposed an alternative theory to predict individual behavior: Theory of Planned Behavior (TPB).

8.2.1.3 Theory of Planned Behavior

TRA asserts that individual behavior is totally controlled by himself (Fishbien and Ajzen, 1975). Eleven years later, in 1986, Ajzen and Madden

underlined the importance of individual feelings to conduct and control an action criticized this statement. Hence the development of TPB proposed by Ajzen (1991). This theory stated that the individual behavior is directly determined by his intention.

TPB is characterized by the introduction “perceived behavioral control” that determines “usage behavior.” Ajzen (1991) stated that behavioral intention is a combination of three variables: (1) attitude, (2) subjective norm, and (3) behavioral control. The later refers to perceptions constraints behavior (Ajzen, 1988). It explains individual’s perception toward difficulty or easy of performing such behavior (Ajzen, 1991).

TPB proves that attitude, subjective norm and perceived behavioral control influence positively the individual Intention to accept a technology. Its main interest is to provide a model that could be used in all situations where the behavior is intentional. The main limit of this theory was absence of factor analysis explaining the link between beliefs and intentions.

8.2.1.4 Social Cognitive Theory

Bandura (1977) developed the Social Cognitive Theory (SCT). He considered contributions of behaviorism and socio-psychology. SCT placed individuals at the center of cognitive, behavioral and contextual factors interaction. It differed from IDT, TRA and TPB that emphasized the influence of environmental factors on the individual behavior. It explains how people acquire and maintain their behaviors by providing a basis for intervention strategies. In 1980, Bundura stated two main objectives of the (SCT):

- To understand and predict individual and groups behavior; and
- To identify methods and ways to change behavior.

For Bandura (1986), human behavior is seen as the product of dynamic interactions between personal and environmental influences. He underlined the importance of interrelation between three factors: personal, environmental and behavioral. Thus the influence of the environment on behavior remains prominent.

Social cognitive theory is related to a set of variables:

- **Social environments:** It includes family members, friends and colleagues. It provides a framework for understanding the behavior;

- **Situation:** It explains environment perception. It refers to mental or cognitive representations that can influence individual behavior;
- **Behavioral capacity:** It includes knowledge and skills to perform a given behavior;
- **Expectations:** They explain the results of proactive behavior;
- **Self-efficacy:** It is defined as a set of beliefs about the ability of a person to adopt such a behavior; and
- **Reciprocal Determinism:** It explains the dynamic interaction of the person behavior and environment.

SCT argues that technology acceptance is not based on convincing individuals of its profits only; it requires their skills and their trust toward this technology.

8.2.1.5 Interpersonal Behaviors Theory

Interpersonal Behavior Theory (IBT) referred to various disciplines (Sociology, Psychology and Anthropology) to understand and predict the individual behavior. It integrates most of variables presented in the behavior theories (TRA and TPB). It introduces two sets of factors: (i) factors related directly to behavior (genetic factors, social status, Habits, Attitudes, Intentions and Behavior); (ii) Factors related to the environment (culture, social factors and facilitating Conditions).

Triandis (1980) referred to principles of the TRA. To develop his model, he integrated “habit” and “External conditions” that facilitate or complicate individual behavior. Furthermore, he distinguishes between “beliefs” that bind emotions to behavior, and those that bind the act to future consequences.

IBT states that individual behavior is determined by three main factors: (i) intention to accept this behavior, (ii) habit and (iii) facilitating Conditions. IBT model showed that Habit influences both of behavior and attitudes (affect), while facilitating conditions include contexts and situations that could foster behavior accept. Intention as the most important variable of this model is determined by individual feeling toward a behavior

(Attitude of Affect), by social factors and by outcomes expected of his behavior (perceived consequences).

IDT, TRA, TPB, SCT and IBT provide a theoretical basis to predict the individual behavior toward Information technology.

8.2.2 TECHNOLOGY ACCEPTANCE MODEL

Researches conducted on IT acceptance (Karahanna, Straub and Chervany, 1999) showed that a large number of models were derived from TRA (Ajzen and Fishbien, 1980). One of these models is the Technology Acceptance Model (TAM) (Davis et al., 1989).

TAM is the most used by researchers in Information Systems field.

It is considered the most robust model to identify variables influencing individual acceptance to use particular technology or not (Venkatesh and Davis, 2000).

It was introduced to explain individual behavior toward computers (Davis et al., 1989). It mobilized TRA to identify key relations between its main constructs: (i) perceived usefulness, (ii) perceived ease of use, (iii) attitude, (iv) intention and (v) usage. It is a causal model that studies relationships between these five constructs. According to TAM, IT use is explained by the Behavioral Intention.

TRA gathered all beliefs in one construct: subjective norms (Ajzen, 1991). TAM enriched TRA by incorporating explicitly external variables to model user behavior. It shows how these variables impact two specific beliefs: PEOU, PU, and how they influence Attitude and Intention to Use Technology to predict actual use.

TAM model suggested that IT use could be determined by behavioral intention. This later is influenced by attitude toward IT usage. PU and PEOU are the prominent components of this model. They were inspired from IDT (Rogers, 1995) that considered complexity and relative advantage as two main determinants of innovation adoption. Complexity joined PEOU, while the Relative Advantage indicated PU.

PU explained the degree to which a person believes that using a particular technology would improve his work performance (Davis, 1989). In other words, it's a subjective probability indicating that an individual

agrees technology usage would improve his performance in an organizational context. Direct relation between PU and Behavioral Intention is based on the assumption that IT use decision is made on the IT use performance perceived by the user. Furthermore, PU is influenced by the PEOU and many other external variables.

PEOU indicated the degree to which the user thinks that the use of a particular technology would be without effort. It implies that a technology perceived as ease is more likely to be used. Benbasat et al. (1986) noted that the PEOU is influenced by characteristics of the technology itself.

These two main constructs, PU and PEOU are affected positively or negatively by external factors (Agarwal and Prasad, 1998, Hong et al., 2002). The model provides a basis for tracing the impact of external factors on internal beliefs, “Attitude” and “Intention”.

In summary, TAM was the most widely model, the most robust and the most parsimonious model used to predict technology acceptance across various contexts (Holden and Karsh, 2010; Egea and González, 2011). It is due to its attribute of being a generic model that can be applied to any context using IT. TAM posits that two fundamental beliefs, PU and PEOU, determined an attitude, behavioral intention (BI), and actual usage of Information Technology (Davis et al., 1989).

8.3 ACCEPTANCE OF E-HEALTH

The ubiquitous usage of Information Technology in healthcare context engendered e-health concept. Referring to Eysenbach (2001), e-health is an emerging field in the intersection of medical informatics, health and business. It is driven from the expansion of health care informatics and biomedical computer (Bennani et al., 2008). In this chapter, E-health is seen as the usage of IT healthcare activities. To understand E-health acceptance, many studies have been developed testing TAM. The following section reviews studies predicting E-health by healthcare professionals for the period between 2007 and 2013.

In 2007, Wu et al., aimed to develop a conceptual framework for significant factors influencing the medical professional behavioral intention to accept Mobile Healthcare Systems (MHS). TAM and IDT served as

the theoretical basis to develop their research model. Confirmation factor analysis was performed to test the reliability and validity of the measurement model, and the structural equation modeling technique was used to evaluate the causal model. Their results indicated that compatibility, PU and PEOU significantly affected healthcare professional intention. Furthermore, MHS self-efficacy had strong indirect impact on healthcare professional acceptance intention through the mediators of perceived usefulness and Perceived Ease of Use. However, the hypotheses for technical support and training effects on the PU and PEOU were not supported.

Kim and Chang (2007) observed the importance of health information websites, and noted the limit studies presenting significant solutions to problems related to effective operations of health information websites. To overcome this limitation, they explored factors affecting health information websites acceptance from the perspective of both the user and the provider. To do, they employed TAM and extend it by four exogenous variables: (i) information search, (ii) usage support, (iii) customization, and (iv) purchase and security. The developed model was tested on a target population of 250 online consumers. Only 228 respondents were retained. Results from the structural equation analysis suggested that PU had a significant effect on Customer Satisfaction. Moreover PEOU turned out to have an indirect effect on this satisfaction. Healthcare services are radically different from ordinary services. Self-Service Technology (SST) is considered as a means to reduce costs and improve quality in the health care sector. Lanseng and Andreassen (2007) examined the introduction of this technology in health diagnosis. For this issue, they mobilized TAM to identify factors that may influence the public's acceptance and adoption of a self-diagnosis system. The study was conducted on a population of 470 inhabitants in an affluent suburb of Oslo the capital of Norway. Results revealed that TAM has an excellent ability to predict SST acceptance in the health care services context. As for Park and Chen (2007), they investigated human motivations affecting medical doctors and nurses' smartphone acceptance. They referred to TAM, users' perceived adoption under the self-efficacy, and innovation attributes to develop their model. To test this later, 823 questionnaires were mailed to medical doctors and nurses, and healthcare providers stationed throughout clinics and the main hospital in the healthcare network. Only 135 were received and 20 were

returned undeliverable. Results showed that medical doctors and nurses intention to accept smartphone was influenced by PU and attitude toward using smartphone.

In 2008, Wu et al., noted that several healthcare organizations invested in adverse event reporting systems to prevent adverse events and medical errors. However, a number of these systems have failed. Hence, the authors, tried to understand factors that might influence healthcare professional's acceptance of adverse event reporting system. To do, they extended TAM by integrating trust and management support to develop their model. After, they tested it on a population of 940 health professionals from 144 Taiwanese hospitals implementing reporting systems. The outcomes indicated that PU, PEOU, subjective norm, and trust had a significant effect on a professional's intention to accept an adverse event reporting system. Changes in health insurance policy engendered operating expenses within Taiwanese hospitals. Hence the latters attempted to and improve their efficiency throughout integration of the electronic logistics information system within department of central warehouse and the nursing stations. Tung et al. (2008) noted that the first step to achieve this integration is to examine the intention of nurses to accept e-logistics information systems. They combined IDT, the main TAM constructs (PU and PEOU), Trust and Perceived Financial Cost to propose hybrid technology acceptance model. After testing this model on 350 nurses, results revealed that compatibility, PU, PEOU, and trust had a great positive influence on the nurses' intention to adopt e-logistics information system.

Carter (2008) observed the importance of physicians' Perceived Risks associated to usage of novel medical technologies. Hence, he attempted to investigate physicians' attitudes toward the use of smart fabric technology. Otherwise, he examined how PEOU and PU affect physicians' attitudes and intention to accept this technology. Empirically, Carter, collected data using a web-based survey instrument delivered via email to a randomly selected group of physicians. The study concerned 2210 physicians, physician's assistants and medical students. Results showed significant effects of attitude on physicians 'intention to accept smart fabric technology. As the web-based learning technology has been widely recognized and become a valuable and legitimate learning tool for health care professionals, Chen et al. (2008) tried to understand its importance for nursing. They tried

to understand public health nurses' intention to accept this technology in pre-implementation stage. They based on TAM constructs to predict this intention. From 369 health centers in Taiwan, 202 public health nurses randomly selected participated in this study. Results of the multiple regression analyzes indicated that PU is the most significant factor for public health nurses' intention to accept web-based learning.

In 2009, Aggelidis and Chatzoglou attempted to test the applicability and effectiveness of technology acceptance models in health care area in Greece. They developed and tested the modify Technology Acceptance Model to examine factors affecting intention of staff (Medical, Nurses and Administrative personnel) using Hospital Information System (HIS) and working in public health institutions. Their results showed that Attitude, PU, Ease of Use, Social Influence, Facilitating Conditions and Self-Efficacy influenced positively personal intention to accept HIS, whereas no significant influence is found between anxiety and this intention. Authors concluded that their modified model can explain 87% of the variance of behavioral intention. Yu and Li (2009) applied a modified Technology Acceptance Model (TAM2) to examine factors explaining Health IT applications by caregivers. Their results revealed that PU, PEOU and computer skills had a strong positive impact on intention of caregivers to accept Health IT. However, Image had the significant negative impact on this intention. Authors pointed out that the research model explained 34% of the variance caregiver's behavioral intention. Lishan et al. (2009) explored differences in perception and acceptance levels of Female-focused Healthcare Applications (FHA). They aimed at detecting factors that would encourage acceptance of these applications. They referred to TAM to develop Female-focused Acceptance Model (FAM). Afterwards, they tested the developed model on a sample comprising of 241 healthcare workers (HC) and 830 non-healthcare (NHC) workers. Results of the regression analyzes showed that PU and PEOU had a significant influence on the intention of female healthcare workers to accept FHA.

Whetstone and Goldsmith (2009) investigated three main questions: (1) what motivates consumers to consider Personal Health Records (PHR) acceptance? (2) What consumer characteristics promote intended PHR usage? (3) Do privacy and security (PS) concern act as impediments to intended PHR usage? Their purpose was to examine factors that bear on

consumers' intention to create and to use (PHRs). They laid on TAM to develop their model. This later was tested on a total of 542 U.S. college students with an online questionnaire. Results found that PU influenced significantly PHR acceptance by consumers. Djasasbi, et al. (2009) examined the acceptance of a specialized telemedicine system for microbiology consultation and diagnostics by microbiology laboratory assistants. They referred to TAM constructs and affect variable to develop their model. This later was tested using a laboratory experiment on Thirty-nine microbiology laboratory assistants. Results of the regression analysis revealed that PEOU might not be so important in the healthcare field. As for PU and attitude they showed a significant positive influence on the intention of microbiology laboratory assistants to accept Healthcare Information System.

In 2010, Veer and Francke stated that a great number of health care organizations are integrating electronic patient records (EPR). This integration implied a change in nurses work routines and allowed quality improvement. They explored determinants of nursing staff attitude towards using EPR by extending the TAM. Empirically, the model was tested on 685 nursing staff members working in Dutch hospitals, psychiatric organizations, care organizations for mentally retarded people, home care organizations and nursing homes or homes for the elderly. One of the relevant results of this study was that attitudes towards EPR were primarily associated with job-related characteristics and PU with respect to care quality.

The low rate of mobile healthcare acceptance in Taiwan pushed Wu et al. (2011) to propose a model to understand this acceptance. For this issue, they referred to TAM and TPB. A survey was conducted to examine mobile acceptance healthcare by health professionals in hospitals. 140 valid questionnaires were retained, representing a response rate of 17.5%. Results indicated that Attitude, PU, Perceived behavioral control and Subjective norm had a significant impact on healthcare professionals to accept mobile healthcare. Egea and González (2011) analyzed physicians' individual acceptance decisions of electronic health care records (EHCR) systems. For this purpose, they referred to the original version of the TAM and extended it by trust and risk-related factors. The final sample consists of 254 physicians with private medical practices located in southern Spain representing response rate of about 18%. Results stressed the special importance of Attitude, Trust and PU in predicting EHCR systems

by physicians. Melas et al. (2011) underlined the qualitative nature and usage of small convenience samples by researches attempting to understand doctors' and nurses' technology acceptance in the workplace. Hence, they mobilized TAM and introduce physicians' specialty as a moderator to develop their research model explaining clinical information systems acceptance. They tested their model in a random sample of 604 medical staff (534 physicians) working in 14 hospitals in Greece. Their results showed that TAM constructs, especially PU, PEOU and Attitude predicted a substantial proportion of the intention of doctors and nurses to accept clinical information systems. H. M. Mohamed et al. (2011) conducted a research to understand population behavior as well as the position of factors affecting e-Health acceptance. To do they employed TAM extended by two sociocultural factors 'trust' and 'tangibility' and proposed e-Health Technology Acceptance Model (e-HTAM). To empirically assess its validity, a pilot study was conducted in UAE and UK. The participants were randomly selected with no prior preferences. The data was collected through online questionnaire. Results revealed the positive correlation between the dependent variable intention and the independent variables: 'trust' and 'tangibility' and the three constructs TAM constructs (PU, PEOU and SN).

The rapid development of healthcare information system attracted the intention of more and more health professionals. By the way, Pai and Huang (2011) proposed a model that predicts acceptance of these systems. The proposed model is inspired from TAM and the Information system success model (DeLone and Mclean, 1992). Empirically, authors conducted questionnaire survey among district hospital nurses, head directors, and other related personnel. Results of the Structural Equation Modeling (SEM) analysis showed that the proposed factors, especially TAM constructs (PU and PEOU) positively influence users' intention to accept a healthcare system.

In 2012, Aldosari underlined the importance of studying archiving and communication systems (PACSs) acceptance by users. He noted that this acceptance is needed given the growing financial investment associated with these systems and its influence on the expected benefits. He used TAM to assess the level of acceptance of the host PACS by staff in the radiology department in Saudi Arabia. He administrated a questionnaire survey of 89 PACS users to collect data. His results found that the

three constructs of PU, PEU, and change explained 41% of the variation in PACS user acceptance.

Asua et al. (2012) conducted a pilot research of a telemonitoring system for the management of chronic patients in primary care in the Bilbao Primary Care Health Region. They aimed to (1) to understand factors related to healthcare professionals' acceptance of the telemonitoring system and (2) to apply an adapted theory-based instrument to assess healthcare professionals acceptance of this system. For these issues they relied on TAM and extended it by Compatibility, and Facilitators. To test their model, they distributed a questionnaire to a total of 605 nurses, general practitioners and pediatricians. Results of the logistic regression analysis indicated that PU predicted significantly intention of health professionals to accept telemonitoring system. Moreover, the extended model was more powerful in predicting this intention.

Yan and Wang (2012) noted the studies limit exploring physician's willingness to accept an e-health care system. Hence, they tried to better understand and predict physicians' acceptance of asthma care mobile service (ACMS) in Taiwan. They based on the TAM and included "subjective norm," "innovativeness," and "managerial support" to predict this acceptance. Of the 700 questionnaires distributed to physicians using ACMS, 504 completed were received. After their analysis by SEM method, results found that attitude, PU, SN and PEOU were the most critical factors affecting physicians intention to accept ACMS. Chen and Hsiao (2012) attempted to better understand factors affecting physician hospital information systems (HIS). To do they referred to TAM and extended it by three quality antecedents (system quality, information quality, and service quality) as the exogenous variables. The model was tested using a survey that targeted physicians with over 1 year's experience using HIS in the selected case hospital. Results of the partial least square approach showed that PU and PEOU significantly affected HIS acceptance by Taiwanese physicians. Moores (2012) referred to TAM to develop an integrated model predicting clinical management system acceptance by hospital workers. They retained PU and PEOU as the fundamental predictors of this acceptance. They targeted clinical management system (CMS) used in a public, regional hospital with approximately 900 clinical staff, which includes physicians, nurses, and allied health workers. To test their

research model, they used the partial least squares (PLS) and SmartPLS. Results showed that in a mandatory use, Attitude, PU and PEOU had no significant influence on system use. Ketikidis et al. (2012) applied a modified version of the TAM to assess the relevant beliefs and acceptance of Healthcare Information Technology (HIT) by medical doctors and nurses systems. They used the two-stage cluster sampling method. The first stage concerned three randomly selected clinics in the city of Skopje. As for the second concerned nurses and medical doctors employed in the selected clinics. Only 133 questionnaires of the 200 distributed were analyzed using multiple linear regression analysis. Results found that PEOU, but not PU, relevance and subjective norms directly predicted health professionals' intention to accept HIT. Jimoh et al. (2012) investigated the potential of Healthcare information technology (HIT) acceptance within maternal and child health workers in rural Nigeria. They aimed to propose a model that could predict IT acceptance in health in sub-Saharan Africa. They referred to TAM to develop their research model. To test the later, a prospective, quantitative survey design was mobilized to collect data from quasi randomly selected clusters of 25 rural health facilities in 5 of the 36 states in Nigeria. Results of the Regression analysis revealed that PU and PEOU were the significant factors influencing HIT acceptance by midwives and health extension workers in rural Nigeria.

In 2013, Cheng attempted to explore nurses' intention to accept e-learning system. Otherwise, he studied the role and the relevance of interaction factors, intrinsic motivators and extrinsic motivators, PU and PEOU in explaining this intention. Empirically, he gathered data from nurses working at two regional hospitals in Taiwan. Analysis of 218 responses by the SEM method showed that effects of PU, PEOU and Flow on nurses' Intention to Use e-learning system were very significant. Chow et al. (2012) pointed out the role of clinical imaging in disease diagnosis and patient care. Moreover, they underlined the need for nurses to acquire skills and knowledge in radiological examination requests and image interpretation. After describing the clinical imaging to facilitate independent learning in image interpretation, they explored factors affecting intentions to accept the portal. They referred to TAM extended by the computer self-efficacy construct as an external variable. The study was conducted on 188 nursing students. Results of the SEM method revealed

that Attitude had a strong effect on this intention, followed by PEOU and self-efficacy. As for Bennani and Oumlil (2013), they attempted to identify factors influencing Health Information Technology (HIT) acceptance by nurses in Morocco. They referred to TAM and extended it by two added constructs “Trust” and “Image.” The two authors conducted their study on a sample of 250 nurses working in health public and private organizations located in Agadir city, south of Morocco. Their results revealed that the proposed model explained 44.1% of total variance in the intention to accept HIT by nurses. Moreover, PU, Attitude, Trust and Image showed significant influence the nurses’ intention to accept HIT.

Table 8.1 synthesises studies mobilized TAM model to predict E-health acceptance by healthcare professionals. Most of these studies were conducted in developed countries (Wu et al., 2007; Lanseng and Andreassen, 2007; Park and Chen, 2007; Carter, 2008; Yu et al., 2009; Asua et al., 2012). Nevertheless, few studies have been conducted in developing countries (Jimoh et al., 2012; Bennani and Oumlil, 2013). The majority of studies concerned nurses acceptance of E-health (Tung et al., 2008; Chen et al., 2008; Veer and Francke, 2010; Asua et al., 2013; Cheng, 2012; Bennani and Oumlil, 2013) followed by physicians (Carter, 2008; Egea and González, 2011; Yang and Wang, 2012; Chen and Hsiao, 2012). Concerning e-health applications, table, shows variation in these applications (Mobile healthcare systems (MHS), electronic patient records (EPR), PACS, personal health records (PHRs)...). Furthermore, the highest variation in predicting E-health acceptance was found in southern Spain (96.4%) where the respondents were only physicians.

8.4 RESEARCH MODEL AND HYPOTHESES

The research model is inspired from TAM (Davis et al., 1989). It includes three principal TAM constructs: Perceived Usefulness, Perceived Ease of Use and Attitude. Two added constructs “Image” and “trust” extended this model. These constructs are considered as external variables supposed to enhance Moroccan healthcare professionals’ intention to accept E-health (**Figure 8 1**). We chose TAM, because it is considered a salient model predicting IT acceptance in various contexts. Moreover, researches referring to TAM to study E-health acceptance are very limited.

TABLE 8.1 Research Works Applying TAM and Its Extensions in E-health Context

Authors	Year	Significant TAM factors	E-health Applications	Population studied	Country	R ²
Wu et al.	2007	PU and PEOU	Mobile healthcare systems (MHS)	Physicians Nurses and Medical technician	Taiwan	0.70
Kim and Chang	2007	PU, PEOU	Health information websites	Online consumers	Korea	NA
Lanseng and Andreassen	2007	PU, PEOU	Self-service technology internet-based medical self-diagnosis	Inhabitant of Oslo	Norway	NA
Park and Chen	2007	PU and Attitude	Smartphone	Doctors and nurses	USA	0.69
Wu et al.	2008	PU, PEOU, SN	Event reporting systems	Healthcare professional	Taiwan	NA
Tung et al.	2008	PU, PEOU	E-logistics information system	Nurses	Taiwan	0.66
Carter	2008	Attitude	Smart fabric medical innovation	Physicians, physician's assistants and medical students	USA	NA
Chen et al.	2008	PU	Web-based learning	Public health Nurses	Taiwan	0.078
Aggelidis and Chatzoglou	2009	Attitude, PEOU, Social	Hospital Information System	Healthcare professional	Greece	0.87
Yu et al.	2009	PU, PEOU, Image	Health IT applications	Healthcare staff	Australia	0.46
Lishan et al.	2009	PU and PEOU	Telemedicine and e-health applications	Female healthcare and non-healthcare workers	Singapore	54–68%
Whetstone and Goldsmith	2009	PU	Personal health records (PHRs)	Students	USA	0.38

TABLE 8.1 Continued

Authors	Year	Significant TAM factors	E-health Applications	Population studied	Country	R²
Djamasbi et al.	2009	PU and Attitude	Healthcare Information System: laboratory telemedicine system	Microbiology laboratory assistants	USA	0.61
Veer and Francke	2010	PU	Electronic patient records (EPR)	Nurses	Netherlands	0.32
Wu et al.	2011	PU	Mobile healthcare	Physicians and nurses	Taiwan	0.63
Egea and González	2011	PU and PEOU	Electronic health care records (EHCR systems)	Physicians	Southern Spain	0.964
Melas et al.	2011	Attitudes PEOU and PU	Clinical information systems (CIS)	Doctors' and nurses	Greece	NA
H. M. Mohamed et al.	2011	PU, PEOU, SN	e-Health	Participants randomly selected	UK and UAE	NA
Pai and Huang	2011	PU and PEOU	Healthcare Information System	Nurses head Directors, and other related personnel	Taiwan	NA
Aldosari	2012	PU, PEOU	PACS	Staff in the radiology department	Saudi Arabia	0.38
Asua et al.	2012	PU	Telemonitoring system for chronic care patients	Nurses	Spain	R1 = 0.63
Yang and Wang	2012	Attitude, PU, PEOU, SN	Asthma care mobile service (ACMS)	Physicians	Taiwan	0.86

TABLE 8.1 Continued

Authors	Year	Significant TAM factors	E-health Applications	Population studied	Country	R ²
Chen and Hsiao	2012	PU and PEOU	Hospital information systems (HIS)	Physician	Taiwan	0.545
Moores	2012	PU	IT	Physicians, Nurses and allied health	France	More than 0.3
Ketikidis et al.	2012	PEOU and SN	Health information technology (HIT)	Nurses and medical doctors	Macedonia	0.679
Cheng	2012	PU, and PEOU	E-learning system	Nurses	Taiwan	NA
Chow et al	2012	PEOU	Clinical imaging portal	Students (nurses)	Hong Kong	0.77
Jimoh et al.	2012	PU and PEOU	ICT	Midwives and health extension workers	Nigeria	NA
Bennani and Oumlil	2013	PU and Attitude	IT	Nurses	Morocco	0.44

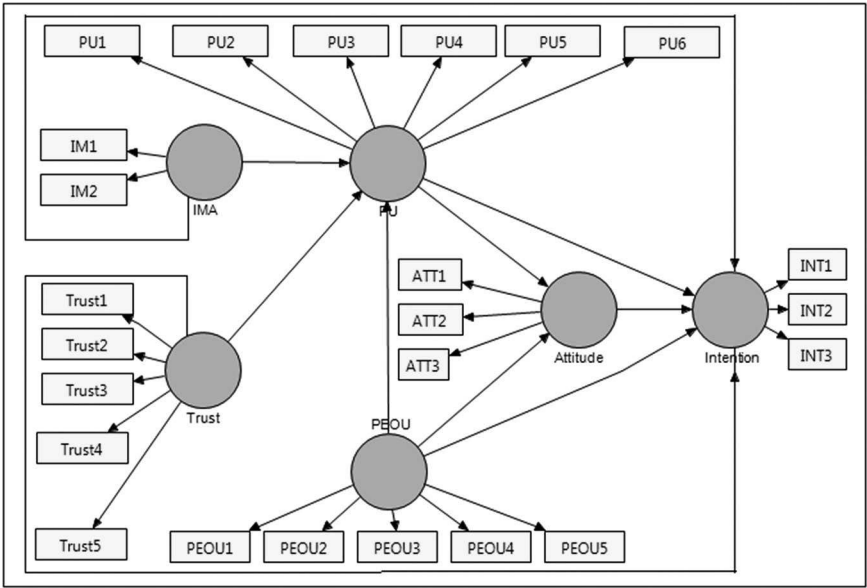


FIGURE 8.1

Perceived usefulness is defined as the degree to which a person believes that using a particular system would enhance his or her job performance. In the Information System area, it means, whether if IT usage can improve user' performance, productivity and efficiency, it will affect his intention to accept it. Therefore, Perceived usefulness is supposed to influence positively directly and indirectly (via attitude) the intention of healthcare professionals to accept E-health. Hence, these relationships are hypothesized as follows:

- H1: Perceived Usefulness is positively and directly related to the Moroccan healthcare professionals' intention to accept E-health.
- H2: Perceived Usefulness is positively and indirectly (through attitude) related to the Moroccan healthcare professionals' intention to accept E-health.

As for Perceived Ease of Use, it refers to the extent to which a person believes that using a particular system would be free of effort. In Information system arena, user intention is affected by his feelings about whether this system is easier to use. Referring to Davis et al. (1989), Perceived Ease

of Use is supposed to influence positively directly and indirectly throughout attitude, the intention of Moroccan healthcare professionals to accept E-health. Therefore, this relationship is hypothesized as follows:

H3: Perceived Ease of Use is positively and directly related to the Moroccan healthcare professionals' intention to accept E-health.

H4: Perceived Ease of Use is positively and indirectly (via Attitude) related to the Moroccan healthcare professionals' intention to accept E-health.

TAM considered Attitude as an important factor predicting behavioral intention. In healthcare context, it is supposed that professionals with a more positive attitude toward IT are likely to be more satisfied with E-health and view it as more beneficial. Therefore, we hypothesized:

H5: Attitude toward E-health is positively related to the Moroccan healthcare professionals' intention to accept E-health.

As mentioned above, an external variable (image) is expected to influence acceptance of E-health by Moroccan healthcare professionals. This construct is drawn from Venkatesh and Davis (2000), and it is due to the need of Moroccan health professionals to improve their image toward other health actors. They thought that using IT could improve their image towards other professionals or even all actors belonging to health arena.

Image is defined as the degree to which usage of an innovation is perceived to enhance one's status in one's social system (Moore and Benbasat, 1991). In the Moroccan healthcare context, professionals with a higher need for social recognition and reputation are likely to accept E-health. Hence we hypothesized:

H6: Image is positively related to the Moroccan healthcare professionals' intention to accept E-health.

H7: Image is positively related to PU.

Trust is defined as the measure of the belief and goodwill that policymakers feel in and for trusted people (Doney et al., 1998). Garbarino and Johnson (1999) added that Trust is the customers' confidence in quality and reliability of the services offered by an organization. Prior studies showed the positive direct relation between Trust and Intention (Gefen et al., 2003). In healthcare context Shibl et al. (2013), Bennani and Oumlil

(2013) pointed out the importance of the trust to predict IT adoption by healthcare professionals. For this chapter it supposed that:

- H8: Trust is positively related to the Moroccan healthcare professionals' intention to accept E-health.
- H9: Trust is positively related to the Perceived usefulness.

8.5 RESEARCH METHODOLOGY

Consistent with previous researches studying IT acceptance by healthcare professionals (Aggelidis and Chatzoglou, 2009), a quantitative method was performed to test the hypotheses above. This study concerned healthcare professionals (physicians and nurses) working in the Provincial Hospital (PH) Hassan Ist of Tiznit city, located in south of Morocco. The choice of this city is due to the fact of belonging to the region considered as the Moroccan leading site for integration E-health.

The public health infrastructure of Tiznit province consist of provincial hospital Hassan Ist and 84 health centers. These latter include 78 rural centers and 6 urban centers. This chapter is limited to the provincial hospital Hassan Ist. This hospital was created in 1981. It has a capacity of 402 beds; It provides basic medicine services. Moreover, it ensures the following specialties: Ophthalmology, dermatology, infectious diseases, Ear Nose and Throat, Psychiatry, Pneumo-phtisiology, Cardiology, Gastroenterology, Rehabi-litation, Neurology, Urology, Endocrinology, Trauma and medical-surgical resuscitation.

Since 2009, the Provincial hospital Hassan Ist developed internal applications by the IT department. These applications covered the following services:

- Admissions Office: Since 2009, this office used software called BAF which deals with entry tickets, exit, billing, death and birth notices.
- Radiology department: It is equipped with "Visual PACS" system that manage results of different units: echography, mammography, Radio and OS-Lung and Scanner.
- Laboratory service: It integrates IS Laboratory system. It displays results of analysis performed by different units, prints them with barcodes.

- **Contracts service:** It contains an information system called “Gide.” This system allows linking the hospital to the provincial treasury.

Data were collected via a survey questionnaire that was divided into two parts. The first one captured perception of model constructs by participants. All the measurements items were anchored on a 7-point Likert scale from “strongly disagree” (1) to “strongly agree” (7) in which respondents indicated an appropriate response. As for the second part, it captured demographic information about each participant. Questionnaires were randomly distributed. Respondents were volunteers asked to complete the questionnaire according to their perception towards IT acceptance to perform their tasks.

The concerned population here was 219 individuals and involved 36 physicians, 138 nurses and 25 administrative staff. Only 160 questionnaires were distributed. This limitation is mainly due to the difficult access to all hospital wards, and to the teams shift. The data collection process took nine weeks, from the middle of June 2013 till the middle of August 2013. This delay is due to three reasons: (1) Low education level of some nurses leads difficulties in understanding the French version of the questionnaire (2) Translation of the questionnaire from French to Arabic or Berber (3) Physicians and nurses were always annoyed with tasks that prevent them to complete the questionnaires.

Of the 160 questionnaires distributed, 125 were completed and retained for the analysis representing a response rate of 78.125%. It consisted of 26 physicians, 71 nurses and 28 administrative staff. Most of the respondents were men (68%) and aged less than 40 years old (55.2%) (Table 8.2). Note that this chapter concerned physicians and nurses only.

8.6 DATA ANALYSIS AND RESULTS

The research model represented in Figure 8.2 was evaluated using Partial Least Squares (PLS) approach. It is classified as a powerful method to study structural models involving multiple constructs with multiple items (Rigdon, 1998).

For this chapter, we chose PLS approach as the main data analysis technique due to the following three reasons:

TABLE 8.2 Demographic Profile of the Sample (N = 125)

Variable	Content	N	(%)
Gender	Men	85	68.0
	Women	40	32.0
Activity	Physicians	26	20.8
	Nurses	71	56.8
	Administrative Staff	28	22.4
Age	20–25 years	9	7.2
	25–40 years	60	48.0
	40–65 years	56	44.8
Education Level	Baccalaureate (Bac) level	21	16.8
	Bac + 1-Bac + 3	63	50.4
	Bac + 4-Bac + 5	15	12
	Bac + 7 and more	26	20.8

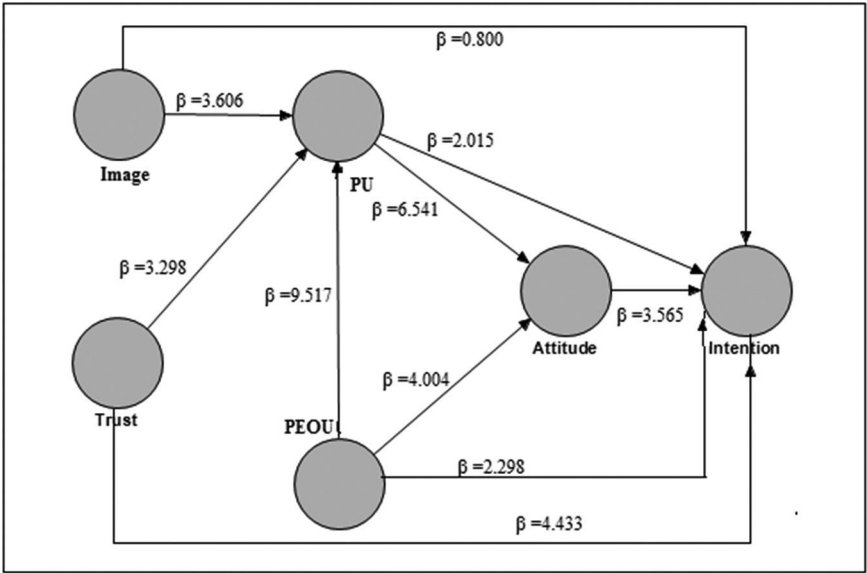


FIGURE 8.2

1. It places minimal demands on sample sizes and data distribution assumptions (Hair et al., 2012);
2. It is a variance-based technique that is oriented towards the predictive aspects of the model (Barclay et al., 1995); and
3. It takes into account the measurement error when testing the structural model (Fornell and Cha, 1994).

However, we should note that PLS method does not provide any global goodness-of fit criterion, which implies a lack of measures for overall model fit (Henseler et al., 2012).

PLS approach consists of two steps to validate predictive models using reflective latent constructs (Henseler et al., 2009). The first step allows the assessment of the measurement model representing the relationships between items and the constructs that they measure. The second step includes the assessment of the structural model depicting the relationships between the constructs as specified by the research model. Authors used SmartPLS software version 2.0M3 to assess both of the measurement and the structural models. They run PLS-Graph using a nonparametric test of significance known as bootstrapping method with 200 resamples to determine the significance levels for loadings, weights, and path coefficients (Gil-Garcia, 2008).

8.6.1 MEASUREMENT MODEL ASSESSMENT

Measurement model called also the outer model. It describes the relationships between constructs and their corresponding manifest indicators. Assessment of the measurement model is performed by: Internal Consistency Reliability (ICR), Convergent validity and discriminant validity (Henseler et al., 2009).

To assess the ICR, we used the Cronbach's Alpha coefficient (α). It is verified when the alpha is above the minimum acceptable threshold of 0.7. In [Table 8.3](#), all constructs indicated composite reliability (CR) above 0.7. Moreover, internal consistency (ICR) of the scales is verified for all constructs because their Cronbach's Alpha doesn't exceed the threshold value and confirmed a satisfactory reliability.

Convergent validity explains the extent to which the indicators of a construct are extremely correlated to each other, and also to indicators of other

TABLE 8.3 Items Loading, Construct Composite Reliability (CR), Internal Consistency Reliability (ICR)

Construct	Items	Item loading	Construct CR	ICR (Cronbach's alpha)
Perceived Usefulness (PU)	PU1	0.834694	0.918462	0.893978
	PU2	0.767659		
	PU3	0.823145		
	PU4	0.839033		
	PU5	0.820333		
	PU6	0.759187		
Perceived Ease of Use (PEOU)	PEOU1	0.688947	0.870839	0.819217
	PEOU2	0.77931		
	PEOU3	0.774069		
	PEOU4	0.775837		
	PEOU5	0.768362		
Attitude (ATT)	ATT1	0.906851	0.866762	0.77026
	ATT2	0.801303		
	ATT3	0.769395		
Image	IMA1	0.919302	0.94197	0.882284
	IMA 2	0.967275		
Trust	Trust 1	0.782884	0.910356	0.877237
	Trust 2	0.862741		
	Trust 3	0.833123		
	Trust 4	0.813936		
	Trust 5	0.798663		
Intention (INT)	INT1	0.909933	0.936178	0.897912
	INT2	0.89988		
	INT3	0.923528		

constructs. Referring to Fornell and Larcker (1981), convergent validity assessment implies to calculate item reliability for each measure, composite reliability for each construct and the average variance extracted (AVE).

Item reliability involves the factor loading as that measure should be 0.70 or greater, which indicates a well-defined structure. As shown in [Table 8.4](#), all of the factor loadings were greater than 0.70, and most of

TABLE 8.4 Diagonal Elements Are the Square Root of the Shared Variance Between the Constructs and Their Measures (AVE); Off-Diagonal Elements Are Correlations Between Constructs (N = 125)

	AVE	Attitude	PEOU	Image	Intention	Trust	PU
Attitude	0.685	0.8279					
PEOU	0.574	0.5877	0.7580				
Image	0.890	0.2297	0.1518	0.9435			
Intention	0.830	0.3938	0.1871	0.0696	0.9111		
Trust	0.670	0.5376	0.4116	0.1436	0.4666	0.8187	
PU	0.652	0.6481	0.6144	0.2556	0.3113	0.4330	0.8079

them were well above that level. Moreover, composite reliability, 0.70 value or higher was recommended by Bacon et al. (1995).

As for AVE, it measures the percentage of variance captured by a construct showing the ratio of the sum of the variance captured by the construct and measurement variance. It is acceptable if it is greater than 0.5.

Concerning discriminant validity, it is assessed following Fornell and Larcker (1981) guidelines. We first use the average variance extracted (AVE), then capture the average variance shared between a construct and its measures. These measures should be greater than the variance shared between the construct and other constructs in the model (Barclay et al., 1995). Moreover, the diagonal elements in the matrix should be significantly greater than the off-diagonal elements in the corresponding rows and columns (Gefen and Straub, 2005). Note, that AVE is generated automatically using the bootstrap technique by the PLS-Graph. Table lists the correlation matrix for the construct. The diagonal elements in the “correlation construct” (in bold) are the square roots of the AVE. Off-diagonal elements are the correlation among construct. Results revealed that AVE for each construct is larger than the correlation of that construct with all the constructs in the model.

Moreover, Discriminant validity is assessed by extracting the factor and cross loadings of all indicators to their respective constructs. Each indicator should be significantly related to the construct, and should not have a stronger connection with another construct (Chin, 2010). Results (Table 8.4) indicated that all indicators loaded on their respective construct are more

highly than on any other, confirming that the constructs are distinct. Hence the discriminate validity was confirmed.

8.6.2 STRUCTURAL MODEL ASSESSMENT

The structural model is also called the inner model. It indicates the causal relationships among constructs in the conceptual model (Hair et al., 2012). It includes estimates of the path coefficients, which indicate the significance of the hypothesized relationship (relationship between the dependent and independent variables). It also provides R^2 value, which determine power of the model (the variance explained by the independent variables). The R^2 value and the path coefficients indicate how well the data support the model (Wixom and Todd, 2005). We run PLS-Graph using a nonparametric test of significance known as bootstrapping method to determine the significance levels for loading weights and path coefficients (Gil-Garcia, 2008). Figure 8.2 shows that the research model could explain 26.3% of total variable (R^2) in the intention of the healthcare professionals to accept E-health. As expected, Perceived Usefulness showed a significant impact on the healthcare professionals intention ($\beta = 2.015$) (H1). Moreover, Perceived Ease of Use had a significant influence on the healthcare professionals intention ($\beta = 2.298$) (H3). Likewise, Attitude depicted a significant positive influence on this intention ($\beta = 3.565$) (H5). Concerning the added constructs, only Trust showed a significant impact on the healthcare professionals intention ($\beta = 4.433$) (H8). However, Image showed no influence on this intention to accept E-health ($\beta = 0.8$) (H6). Results also revealed the significant impact of PU ($\beta = 6.541$) and PEOU ($\beta = 4.004$) on the Attitude confirming H2 and H4. Furthermore, Trust ($\beta = 3.298$) and PU ($\beta = 3.606$) showed a significant influence on PU, confirming H9 and H7.

8.7 DISCUSSION

The main objective of this chapter is to identify factors predicting E-health acceptance by Moroccan professionals working at the provincial hospital Hassan Ist located at Tiznit city, south of Morocco. Technology acceptance model extended by Image and trust constructs was adopted for this issue.

As indicated in results above, Perceived Usefulness influences positively the intention of Moroccan health professionals to accept E-health. This result confirms the pragmatic behavior of the interviewed healthcare professionals, and tends to focus on the usefulness of the information technology itself. Moreover, positive impact of PU on the Moroccan healthcare professionals intention to accept E-health confirms beliefs in using a particular system that would enhance their job performance. Hence, Moroccan health professionals accept this technology whenever it demonstrates its ability to meet their needs. This finding is similar to those of (Wu et al., 2007; Kim and Chang, 2007; Lanseng and Andreassen, 2007; Park and Chen, 2007; Tung et al., 2008; Yu and Li, 2009; Whetstone and Goldsmith, 2009; Veer and Francke, 2010; Egea and González, 2011; Pai and Huang, 2011; Melas et al., 2011; Aldosari, 2012; Jimoh et al., 2012; Yang and Wang 2012; Bennani and Oumlil, 2013).

In agreement with Wu et al. (2007), Kim and Chang (2007), Lanseng and Andreassen (2007), Park and Chen (2007), Tung et al. (2008), Yu and Li (2009), Egea and González (2011), Melas et al. (2011), Pai and Huang (2011), Yang and Wang (2012), Chow et al. (2012) the Perceived Easy of Use influences positively the intention of health professionals to accept E-health. This may suggest that they are ready to use E-health whenever it would be free of effort. Otherwise, these professionals will use technologies if they are relatively effortless and less complicated.

Furthermore, our research showed that the added construct “image” does not influence positively and significantly the intention of the healthcare professionals to accept E-health. Hence, using E-health would not be an opportunity to improve their Image toward other health professionals.

Prior studies stressed the positive direct relation between Attitude and Behavioral intention to accept E-health by healthcare professionals (Park and Chen, 2007; Aggelidis and Chatzoglou, 2009; Gaylin et al., 2010; Melas et al., 2011; Yang and Wang, 2012). Similarly, our research showed that attitude was significant in the Moroccan context. This emphasizes the importance of health professionals’ attitude and their beliefs in the benefits of using E-health.

Concerning Trust, results showed its significant impact on the intention of health professionals to accept E-health. This explained that these professionals’ confidence in quality and reliability of the services provided by E-health would enhance their intention to accept this technology.

In order to more explain E-health acceptance by Moroccan professionals, a comparison considering type of activity was performed. Results revealed that: contrary to nurses ($R^2 = 28.3\%$) physicians showed a high variance value in Behavioral intention ($R^2 = 71\%$). Moreover, Attitude, PU, PEOU, Trust and Image impacted significantly E-health acceptance for physicians. However, Attitude, PEOU and Image were the only main construct determining this acceptance for nurses.

8.8 CONCLUSION

This chapter aimed to identify factors predicting E-health acceptance by healthcare professional in developing countries. The study was conducted in the provincial hospital Hassan Ist, located in Tiznit south of Morocco. It studied intention of physicians and nurses toward acceptance of E-health. Results showed that Perceived Usefulness, Perceived Ease of Use, Attitude and trust explained this acceptance, while Image showed no significant effect.

This study proposed a valuable tool to managers and decision makers working for healthcare organizations in developing countries. It helps them to identify factors determining E-health adoption and ensuring its success within these organizations.

The sample size represents the main limitation to this research work and could bias results. Future work should consider a more accurate representation of the sample, and give more intention to specific constructs such as 'culture' that could help predicting E-health in developing countries.

KEYWORDS

- **developing countries**
- **e-health**
- **health professionals**
- **healthcare organizations**
- **IT acceptance**
- **morocco**
- **Technology Acceptance Model (TAM)**

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